Probabilistic Sizing of Energy Storage Systems for Reliability and Frequency Security in Wind-Rich Power Grids

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***Abstract*—The penetration of wind energy has increased significantly in the power grid in recent times. Although wind is abundant, environment-friendly, and cheap, it is variable in nature and does not contribute to system inertia as much as conventional synchronous generators. Coupled with the low inertia contribution, the generation intermittency of wind power leads to reliability and stability issues in the power system. Energy storage systems (ESSs) are among the most prominent alternatives to alleviate these concerns associated with high wind penetration. This paper proposes a planning strategy to size ESS for the reliability and frequency security of wind-rich power grids. A probabilistic methodology for ESS sizing is developed utilizing a composite reliability-based framework with sequential Monte Carlo simulation (MCS). The MCS generates composite reliability indices for the power system, which are employed to obtain the capacity for a reliability energy storage system (RESS). Simultaneously, the MCS-derived probability of synchronization of conventional generators is integrated into an analytical approach for sizing a frequency support energy storage system (FESS). The effect of wind farm dispersion across geographical regions is incorporated in the framework to study possible reductions in the ESS size while maintaining the system reliability and frequency security. The efficacy of the proposed strategy is demonstrated on the RTS-GMLC test system.**

***Index Terms*—energy storage systems, frequency security, reliability, sizing, wind energy**

I. Introduction

Wind energy is among the fastest-growing renewable energy resources used today for electric power generation. Sustained policy support and economic incentives for wind power generation worldwide have led to its significant growth in recent times [1]. In 2022, wind power production experienced an unprecedented surge, with a remarkable increase of 265 TWh, marking a 14% growth and reaching a total of over 2,100 TWh [2]. This growth rate was the second highest among all renewable energy sources, trailing behind only solar photovoltaics. Nevertheless, to align with the net zero emissions by 2050, which anticipates a wind electricity generation of approximately 7,400 TWh by 2030, it is necessary to boost the annual growth rate of wind generation to around 17% [2]. However, this increasing penetration of wind energy leads to reduced reliability and stability of the power grid due to its variable nature, non-dispatchable characteristics, and displacement of high-inertia conventional sources [3]. Energy storage systems (ESS) have been extensively researched as a solution to mitigate generation variability associated with wind and other renewables while also providing ancillary services for system stability. However, there are still some reservations about deploying ESS for grid-scale applications due to high capital and operation costs. As a result, the sizing of ESS has become an integral component of planning problems in wind-rich power grids.

The literature extensively discusses the impacts of wind generation intermittency on the reliability of power systems [4], [5]. These drawbacks associated with the fluctuating nature of wind power can be alleviated through various approaches, such as the deployment of ESS, the aggregation of wind energy from diverse geographical locations, and flexible load management. While the management of flexible loads could prove to be the most cost-effective approach, their penetration in existing grids is still not significant enough to contribute to system reliability [6]. Among these methods, previous research has extensively focused on the deployment of ESS and resource aggregation to mitigate the variability of wind generation and improve system reliability. Authors in [7]–[9] propose deterministic approaches to size ESS for the reliability of networked microgrids. While the authors develop optimization frameworks to generate optimal ESS sizes, they do not incorporate the effects of stochasticity presented by distributed generation. To account for the stochasticity of renewables, authors in [10], [11] propose planning frameworks for ESS sizing while considering system reliability. However, the number of renewable scenarios used in these frameworks are limited and hence do not capture the full range of uncertainty displayed by these resources. Joint planning frameworks in [12], [13] alleviate this issue by focusing on modeling uncertainties introduced by distributed generation while sizing ESSs. While the joint planning frameworks can simultaneously incorporate the uncertainties of wind generation and economic sizing of ESSs, they do not consider system reliability as an ESS sizing criterion. On the other hand, references [14], [15] have presented probabilistic approaches that use a combination of wind power aggregation and ESS sizing to ensure reliability in wind-rich grids. However, these approaches are limited to reliability applications only and do not consider the frequency support capabilities of the ESS.

In addition to the concerns over system reliability, the non-dispatchable and intermittent characteristics of wind power lead to a drastic reduction in the system frequency response. While some recent works have incorporated the virtual inertia and primary frequency response using virtual synchronous generators and de-loading techniques [16], [17], they are still insufficient in providing the required frequency support to ensure grid code compliance [18]. As of date, ESS remain the prime candidates to supplement wind power integration using their virtual inertia capabilities [19]. Since ESS are fast-acting devices with high ramping capabilities, they are well-suited for providing virtual inertia to the system. For this purpose, ESS need to be sized accurately, depending on the frequency security requirements. References [20], [21] develop sizing algorithms for the ESS to enhance inertial response (IR) and primary frequency response (PFR) in the presence of wind power. However, these works do not employ a probabilistic approach for ESS sizing, so they do not account for the random outages of system components causing loss of system inertia. While sizing the ESSs, references [18], [22], [23] have proposed probabilistic frameworks to incorporate the system inertia loss resulting from the failures of conventional generators. However, all the probabilistic approaches mentioned focus only on IR support of ESS and ignore the primary frequency response (PFR) support in their sizing algorithms, leading to an underestimation of ESS sizes. In addition, none of the works mentioned above consider the alternate methods to reduce the ESS sizes while retaining the same level of frequency security. Since the ESS are expensive in both investment and operation stages, it is important to consider other methods, such as the geographical distribution of wind farms for ESS sizing methods.

In this paper, an improved framework for ESS sizing is presented that is applicable for ESS providing both reliability and frequency support in wind-rich systems. A probabilistic approach based on the sequential Monte Carlo simulation (MCS) is developed, which incorporates the stochasticity and geographical distribution of wind farms, forced outages of generating units and transmission lines in the sizing problem. The system composite reliability indices obtained from the MCS are used to determine the size of reliability ESS (RESS). At the same time, the probability of synchronization of conventional generators obtained from the MCS is deployed in a sizing algorithm for frequency support ESS (FESS). The proposed work also focuses on the potential reduction of ESS size requirements by improving wind power capacity value from the geographical dispersion of wind farms. The major contributions of the paper are summarized as follows:

- Developing an improved probabilistic sizing tool that simultaneously determines storage sizes for reliability and frequency security of systems with high wind penetration. The method is computationally efficient as it does not require different simulations for the sizing of RESS and FESS separately.
- Proposing a single MCS framework for sizing of both RESS and FESS that incorporates uncertainty in wind generation, forced outages of system components, and system operation constraints. Analytical approaches for the sizing of RESS and FESS in the literature do not consider these factors.
- Improving the FESS sizing algorithm to incorporate both virtual inertia and droop response of ESS within an analytical multi-machine load frequency control (LFC) model. The resulting analytical model can easily be incorporated with the proposed optimization-based FESS sizing tool.
- Proposing and studying the potential benefits of wind farm locational diversification in improving system reliability and reducing the storage size requirements.

The remainder of this paper is organized as follows. Section II introduces the models for wind speed and geographical dispersion of wind power. In Section III, the base sequential MCS framework with the DC optimal power flow (DC-OPF) is developed. Section IV and V highlight the methodology for RESS and FESS sizing, respectively. The simulation and results are presented in section VI, demonstrating the ESS sizing and wind aggregation benefits for reliability and frequency security. Section VII concludes the paper with some remarks.

II. Modeling the Wind Power Stochasticity and Locational Distribution of Wind Farms

In this work, the ESS sizes for reliability and frequency stability depend on the system reliability indices and generator synchronization probabilities obtained using a Monte Carlo simulation. Accurate and computationally feasible models for wind speed stochasticity are required to obtain the system reliability indices and the generator synchronization probability under wind integration. This section discusses the wind speed model based on the discrete Markov process and the concept of wind power dispersion and its possible benefits.

A. Modeling Wind Speed using the Discrete Markov Model

To incorporate the impact of wind variability into the MCS, the wind speeds are approximated using a discrete Markov process. The process involves the formation of Markov chains with a finite number of clustered wind states. An illustration of the wind speed model, characterized by n states, is depicted in Fig. 1. In this paper, the wind speed corresponding to a Markov state is determined using the fuzzy C-means clustering algorithm. The Markov model captures the characteristics of clustered wind speeds using state probabilities, occurrence frequencies, and durations. The main assumption of this model is that it assumes the wind speed remains statistically stationary. The Markov model for wind speed has been used in reliability assessment methods due to their computational efficiency and reasonable accuracy for hourly-sampled wind speeds [24]. The state probabilities and transition rates of the wind speed states can be obtained as follows:

$$\rho_{i,j} = \frac{t_{ij}}{D_i}, \quad \mathcal{P}_i = \frac{\sum_{j=1}^{\mathrm{n}} t_{ij}}{\sum_{k=1}^{\mathrm{n}} \sum_{j=1}^{\mathrm{n}} t_{kj}} \tag{1}$$

where n is the total number of clustered wind speed states, t_{ij} are the number of transitions from state i to state j of wind speed state, D_i is the duration of state i wind speed respectively, $\rho_{i,j}$ is the transition rate from state i to state j of wind speed, and $\mathcal{P}_i$ is the state probability for state i.

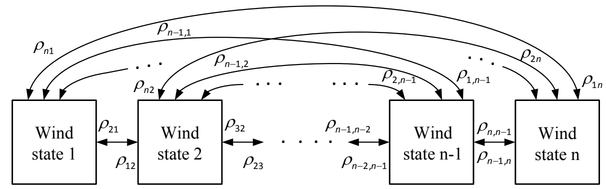


Fig. 1: Wind speed model with n states.

In this work, discrete wind speed states and transition matrices are formulated depending on the location of wind farms and their historical wind data. Using a deterministic wind power output model [5], the output power of a wind farm is calculated. To model the stochasticity in wind farm output, a next-event-based method is used within the Monte Carlo simulation framework in section III to model the switching between the wind speed states for each wind farm.

B. Geographical Dispersion of Wind Power

The major merit of the dispersion of wind power is the reduction in the global output variance of wind power, which can effectively help improve the capacity value of wind farms. Mathematically, the relation between the dispersion of wind farms and the variance in total power output from wind farms spread across geographical regions under certain conditions can help validate the theory. Assuming there are a total of K wind farms, each experiencing distinct wind speeds and patterns. Each of these wind farms is equipped with identical wind turbines, and the power output of the k^{th} wind farm is denoted as P_k^w. The combined power output from all K wind farms can be formulated as follows:

$$P_w^{Total} = \sum_{k=1}^{K} P_k^w \tag{2}$$

The global variance in wind power is then expressed as [25]:

$$\mathrm{var}[P_w^{Total}] = \sum_{k=1}^{K} \mathrm{var}[P_k^w] + 2\left[\sum_{k<l} \mathrm{cov}[P_k^w, P_l^w]\right] \tag{3}$$

where $\mathrm{var}[\cdot]$ and $\mathrm{cov}[\cdot]$ represent the variance and covariance operators respectively. The Pearson's correlation coefficient for the i^{th} and j^{th} wind farms is expressed as:

$$\mathrm{corr}[P_k^w, P_l^w] = \frac{\mathrm{cov}[P_k^w, P_l^w]}{\sqrt{\mathrm{var}[P_k^w]\mathrm{var}[P_l^w]}} \tag{4}$$

Expressing (3) in terms of the correlation coefficient from (4), the resulting expression for the global wind power variance is expressed as follows:

$$\begin{aligned}\mathrm{var}[P_w^{Total}] = &\sum_{k=1}^{K} \mathrm{var}[P_k^w] \\ &+ 2\left[\sum_{k<l} \mathrm{corr}[P_k^w, P_l^w]\sqrt{\mathrm{var}[P_k^w]\mathrm{var}[P_l^w]}\right]\end{aligned} \tag{5}$$

Equation (5) implies that dispersing the wind power among locations with small correlations helps reduce the global output variance of wind power. However, this model of geographical dispersion of wind farms assumes that all turbines within a wind farm are subject to the same wind speeds. This assumption is reasonable for this work as it models the wind speed states in an hourly resolution [25].

An example is used to illustrate the positive effect of wind power dispersion. Supposing two cases, where in case 1, n_w wind turbines are concentrated in one location with output variance per turbine $\mathrm{var}[P_1^w] = \sigma$, the global output variance is expressed as follows:

$$\mathrm{var}[P_w^{Total}] = \mathrm{var}\left[\sum_{k=1}^{1} n_w P_k^w\right] = n_w^2 \sigma \tag{6}$$

In the second case, wind power is distributed equally among two locations with identical individual variance $\mathrm{var}[P_1^w] = \mathrm{var}[P_2^w] = \sigma$ and correlation $\mathrm{corr}(1,2) < 1$. The global output variance in this case is:

$$\begin{aligned}\mathrm{var}[P_w^{Total}] &= \sum_{k=1}^{2} \mathrm{var}[\frac{n_w}{2} P_k^w] + 2 \times \mathrm{cov}[P_1^w, P_2^w] \\ &= \frac{n_w^2\sigma}{4} + \frac{n_w^2\sigma}{4} + 2 \times \mathrm{corr}(1,2) \times \frac{n_w^2\sigma}{4} \\ &= \frac{n_w^2\sigma}{2}(1 + \mathrm{corr}(1,2))\end{aligned} \tag{7}$$

The above example shows that the geographical separation of wind farms reduces the global wind power variance, depending on their correlations. Several experimental studies have reported an improvement in capacity value when wind farms are dispersed over larger distances [26], [27], primarily attributing this improvement to the reduced correlations between wind speeds of diverse locations. Consequently, geographical dispersion of wind farms reduces the size requirement of RESS, as improving the capacity value of wind farms promotes system reliability. At the same time, however, the improved wind power output by geographical dispersion can cause higher displacement of conventional generators, increasing the sizing requirement of the FESS. Studying the effects of geographical dispersion of wind farms on ESS sizes is one of the major contributions of this work.

III. Sequential Monte Carlo simulation for sizing of Energy Storage Systems

The mixed timing sequential MCS proposed in reference [28] is modified in this study to determine the parameters for the sizing of RESS and FESS. The synchronous component of the proposed MCS incorporates the variation in system load, wind, and photovoltaic (PV) farms. In contrast, the asynchronous component changes the states of conventional generators and transmission lines. The steps of the MCS proposed in this work are outlined as follows:

1) Obtain the failure and repair rates for generators and transmission lines. Set the state of each component to available (up) and initialize the wind states for all locations.
2) Initialize the index for the sample path as $s = 1$.
3) Initialize the hourly index $t = 1$.
4) Determine the time for the conventional generators and transmission lines to change their states. The time until the next event, denoted as T_i, for the i-th component is calculated as $T_i = -\frac{1}{\lambda_i} ln(U_i)$ where U_i represents a random number generated from the uniform distribution, λ_i represents the failure rate if the i^{th} component is in up state, or the repair rate when if is in $down$ state.
5) From the vector of transition time for the components, select the shortest time,

$$t_{min} = \min(T_1, T_2, \ldots, T_{G+L}) \tag{8}$$

where G and L denote the system's total number of conventional generators and transmission lines. The t_{min} value indicates when the component changes states from up to $down$ or vice versa.

6) Decrease t_{min} by one hour. If $t_{min} = 0$, goto step 5.
7) Using the current state of wind farms and the transition rate matrices, determine the next state for all wind farms in the system. For example, assuming a wind farm w is currently at wind speed state i among the n states in Fig. 1, a vector of transition times is generated as follows:
$$T_i^w = [T_{i,1}^w, T_{i,2}^w, \dots, T_{i,\mathrm{n}}^w] \tag{9}$$
where,
$$T_{i,j}^w = -\frac{1}{\rho_{i,j}} ln(U_j) \tag{10}$$
where U_j is a random number generated from the uniform distribution and, $\rho_{i,j}$ is the (i,j) component of the transition rate matrix obtained using (1). The smallest element in T_i^w gives the next state of the wind farm among n states.
8) For the current hour, run a DC optimal power flow (DC-OPF) to generate an appropriate dispatch where the load curtailment is minimized and units with low fuel costs are dispatched. Check if the load is curtailed at any bus. If true, record the event and update the system reliability indices. The events are used to determine the RESS size.
9) For the current hour, record the dispatch status of generators. The dispatch status for any conventional generator g is expressed as follows:
$$\gamma_g(t) = \begin{cases} 1 & \text{if } P_g(t) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{11}$$
10) Is $t > 8760$?
   a) If false, goto step 7.
   b) If true, calculate the probability of generator synchronization for sample path s as follows:
$$\mathcal{P}_g(s) = \frac{1}{8760} \sum_{t=1}^{8760} \gamma_g(t) \tag{12}$$
   Check for convergence. If converged, terminate the simulation. Else, $s = s + 1$ and go to step 2.

The steps are repeated until the simulation converges to some prespecified convergence criterion. At the end of the MCS, the reliability indices are used to determine the power and energy ratings for the RESS while the probabilities of generator synchronization are used to size the FESS.

As mentioned in step 8 of the MCS, at every hour t, a DC-OPF is solved to generate a dispatch while minimizing the system load curtailment. In addition to minimizing the load curtailment, the dispatch should consider dispatching generators with low fuel costs for conventional generators and maximizing RER integration while respecting the power grid operational constraints. The DC-OPF is structured as follows:

1) Objective function:
$$\min \left(\sum_{b=1}^{\mathcal{B}} \lambda C_b(t) + \sum_{g=1}^{G} c_g P_g(t) \right) \tag{13}$$
where $\mathcal{B}, G$ represent the overall number of buses and conventional generators within the system, $C_b(t)$ denotes the curtailed load at bus b for an hour t, and λ is a large value to heavily penalize load curtailment in the objective function. G denotes the total number of conventional generators in the system, c_g is the fuel cost (\$/MW), and $P_g(t)$ is the active power generation of generator g. In this formulation, generation from conventional generators is penalized by fuel costs to encourage the integration of renewables into the system. The constraints are expressed as follows:

2) Power balance: For power balance, the power injected into bus b must equal the real power discharged from b. The power balance constraint can be represented as follows:
$$\sum_{g=1}^{G_b} P_g(t) + \sum_{p=1}^{P_b} P_p(t) + \sum_{w=1}^{W_b} P_w(t) + C_b(t) - P_b^L(t) = \sum_{i=1}^{\mathcal{B}} B_{bi}\delta_i(t) \tag{14}$$
where G_b, P_b, W_b denote the total number of conventional generators, PV farms and wind farms connected to bus b. $P_p(t)$ and $P_w(t)$ are the power generated by PV and wind farms, respectively. $P_b^L(t)$ is the demand at bus b, B_{bi} is the bus admittance matrix's imaginary component, and $\delta_i(t)$ is the voltage angle.

3) Transmission line limits: Active power in the transmission lines must be within limits and it is expressed as follows:
$$\left| \frac{\delta_i(t) - \delta_b(t)}{x_{ib}} \right| \le P_{ib}^T \tag{15}$$
where x_{ib} is the reactance between buses i and b, and P_{ib}^T is the maximum transmission capacity of the line connecting buses i and b.

4) Generation limits: The active power output of conventional generators and wind farms must adhere to specific limits, which can be expressed as follows:
$$P_g^{\min} \le P_g(t) \le P_g^{\max} \tag{16}$$
$$P_w^{\min} \le P_w(t) \le P_w(t)^{\max} \tag{17}$$
$$P_p^{\min} \le P_p(t) \le P_p(t)^{\max} \tag{18}$$
where $P_g^{\min}, P_w^{\min}, P_p^{\min}$ denotes the lower limits of conventional generators, wind, and PV farms respectively, $P_g^{\max}, P_w(t)^{\max}, P_p(t)^{\max}$ are the upper generation limits. The PV and wind farm generation upper limit vary along with the hourly variation in input solar irradiance and wind speed respectively. Also, the component status during the MCS hour affects the conventional generator power limits and transmission line limits. For example, if the generator g's state changes to down during hour t, $P_g(t)$ is set to zero.

IV. SIZING OF ESS FOR SYSTEM RELIABILITY

All the load curtailment events are initially extracted from the MCS in the last section. The power rating of the RESS is determined using the expected value of load curtailment during events. Then, the system loss of load probability (or unavailability) and mean down time (MDT) are calculated using these events. The ratio of the unavailability obtained from the MCS and a desired reliability target are then used to determine the unavailability ratio. The time rating of the RESS is derived using the unavailability ratio and the MDT. In addition, a RESS deployment strategy is formulated in this section to deploy the sized RESS back into the system.

A. Sizing of Reliability ESS

The power rating for the RESS is the expected unserved demand during events, and it is expressed as follows:

$$P^{\text{RESS}} = \mathbb{E}\left[\frac{1}{N^{cy}}\sum_{i=1}^{N^{cy}} \mathcal{C}_i\right] \tag{19}$$

where N^{cy} is the total ON/OFF cycles simulated in one sample path and $\mathcal{C}_i$ is the total load curtailment for cycle i.

For the time rating of the RESS, the following approach is proposed. Assume that the system's current availability obtained using the MCS is A_0, which needs to be increased to A_1 using the RESS. Defining the unavailability ratio:

$$\alpha = \frac{1-A_1}{1-A_0} \tag{20}$$

To increase the system's availability from A_0 to A_1, the RESS must be capable of supplying P^{RESS} for a certain time t^{RESS}. The following integral proposed in [29] can be used to determine the value of t^{RESS}:

$$\int_{t^{\text{RESS}}}^{\infty} f_R(r)dr = \alpha \tag{21}$$

where r is the MDT of the system and $F_R(r)$ is the probability density function of the MDT. The following relation is used to determine the MDT [28]:

$$r = \frac{LOLP}{LOLF} \tag{22}$$

where,

$$LOLP = \mathbb{E}\{\Pi\}, \quad \Pi = \frac{1}{T}\sum_{i=1}^{N^{cy}} T_i^{dn} \tag{23}$$

$$LOLF = \mathbb{E}\{\Phi\}, \quad \Phi = \frac{N^{cy}}{T} \tag{24}$$

where T denotes the total simulation time in a sample path, T_i^{up} is the service period and T_i^{dn} is the interruption period. The solution of (21) can be expressed as follows:

$$t^{\text{RESS}} = \frac{-r\ln(\alpha)}{a^R} \tag{25}$$

where a^R is the availability of the RESS. In the real world, the ESS can encounter failures and it needs to be incorporated in the time rating to ensure sufficient storage, which is done using a^R. Finally, the energy rating for the reliability ESS can be obtained as $S^{\text{RESS}} = P^{\text{RESS}} t^{\text{RESS}}$.

B. Deployment of the Reliability ESS

After sizing the RESS, they must be deployed in the system and the composite reliability should be reevaluated. Depending on the obtained sizes, the RESS can be deployed on a single bus or distributed among multiple buses. In the sequential MCS framework, the DC-OPF framework is modified to accommodate the RESS. In the presence of the RESSs, the objective function (13) is modified as follows:

$$\min\left(\sum_{b=1}^{\mathcal{B}} \lambda C_b(t) + \sum_{g=1}^{G} c_g P_g(t) + \sum_{e=1}^{E} c_e^+ P_e^+(t) + c_e^- P_e^-(t)\right) \tag{26}$$

where E denotes the total number of RESSs, $P_e^+(t), P_e^-(t)$ are the charging and discharging power of RESS e. The charging and discharging costs c^+, c^- are introduced to ensure that RESS charge during regular hours and discharge if generation is insufficient to account for the load. These costs are set such that they act as complementary constraints to prevent the simultaneous charging and discharging of RESSs, preventing the need for binary variables in the optimization. Similarly, the power balance constraint is updated as follows:

$$\sum_{g=1}^{G_b} P_g(t) + \sum_{p=1}^{P_b} P_p(t) + \sum_{w=1}^{W_b} P_w(t) + \sum_{e=1}^{E_b} \big(P_e^+(t) + P_e^-(t)\big) + C_b(t) - P_b^L(t) = \sum_{i=1}^{\mathcal{B}} B_{bi}\delta_i(t) \tag{27}$$

The operation of the RESS is subject to the charging, discharging power limits, and state-of-charge (SoC) constraints. The RESS assist the system in maintaining the generation-load balance. For the hourly-operating ESSs, the charging and discharging power $P_e^+(t), P_e^-(t)$ are optimization variables, and they must be within limits as follows:

$$\overline{P_e^+} \le P_e^+(t) \le 0 \tag{28}$$

$$0 \le P_e^-(t) \le \overline{P_e^-} \tag{29}$$

where $\overline{P_e^+}, \overline{P_e^-}$ are the maximum charging and discharging power. Also, the battery SoC must be within limits:

$$\underline{SoC}_e \le SoC_e(t-1) - \frac{1}{E_e^{\text{tot}}}\left(P_e^-(t)\eta_e^- + \frac{P_e^+(t)}{\eta_e^+}\right) \le \overline{SoC}_e \tag{30}$$

where $\underline{SoC}_e, \overline{SoC}_e$ are the lower and upper SoC limits, $SoC_e(t-1)$ is the past hour SoC, η_e^+, η_e^- are the charging and discharging efficiencies, and E_e^{tot} is the energy capacity.

V. Sizing of ESS for System Frequency Security

The first step for FESS sizing is also the MCS in section III, except the generator synchronization probabilities are used instead of the reliability indices. Then, using the synchronization probabilities, the expected values of system inertia and other frequency security parameters are obtained. The frequency security constraints are then developed using these parameters and the proposed multi-machine load frequency control (LFC) model. A procedure is developed to obtain the virtual inertia and droop constants of FESS using the IR and PFR control loops. Finally, the frequency security constraints and FESS virtual inertia, droop constants are incorporated in an optimization framework for FESS sizing. The FESS sizing procedure is described in detail as follows:

A. Derivation of Frequency Security Constraints

The LFC model proposed in [30] is modified to include the FESS and illustrated in Fig. 2. The notations for the governor and turbine parameters of the machines and the FESS parameters are as follows:

- H_g, H^{FESS} is the inertia of g^{th} generator and virtual inertia of FESS respectively.

- D_g is the damping coefficient of generator.
- K_g is the generator droop gain.
- R_g, R^{FESS} is the droop constant of generator and FESS.
- T_g, T^{FESS} is the time constant of i^{th} generator and FESS.
- F_g is the fraction of power from high-pressure turbines of the i^{th} generator.

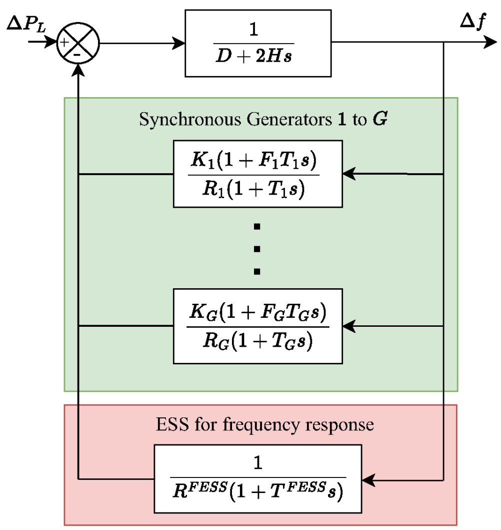


Fig. 2: LFC model for a multi-machine system with FESS.

Using the system LFC model in Fig. 2, the system frequency under a step disturbance ΔP_L is expressed as follows:

$$\Delta f(s) = \frac{\Delta P_L/s}{D + 2Hs + \sum_{g=1}^{G} \frac{K_g(1+F_iT_gs)}{R_g(1+T_gs)} + \frac{1}{R^{\text{FESS}}(1+T^{\text{FESS}}s)}} \tag{31}$$

where,

$$D = \mathbb{E}\{D\} = \sum_{g=1}^{G} \frac{\mathcal{P}_g D_g P_g^{\max}}{P_{sys}}, \tag{32}$$

$$H = \frac{1}{P_{sys}} \left(\underbrace{\sum_{g=1}^{G} \mathcal{P}_g H_g P_g^{max}}_{\mathbb{E}\{H\}} + H^{\text{FESS}} P^{\text{FESS}} \right) \tag{33}$$

In the equations above, P_{sys} denotes the base system MVA. It should be noted that the inertia contribution of a synchronous generator is dependent on the probability of generator synchronization $\mathcal{P}_g$ obtained using the MCS. Since the FESS does not have a damping coefficient, the value of equivalent damping D equals the expected damping value from the MCS $\mathbb{E}\{D\}$. However, the equivalent system inertia is the sum of the expected inertia from the MCS $\mathbb{E}\{H\}$ and the virtual inertia from the FESS. The IR contribution from the FESS depends on the inertia constant and the active power rating.

The frequency deviation after using the inverse Laplace transformation on (31) can then be expressed as follows:

$$\Delta f(t) = \frac{\Delta P_L}{2HT_r\omega_n^2} \left(1 - \frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \cos(\omega_n \sqrt{1-\xi^2} t - \phi) \right) + \frac{\Delta P}{2H\omega_n\sqrt{1-\xi^2}} e^{-\xi\omega t} \cos(\omega_n\sqrt{1-\xi^2} t) \tag{34}$$

where,

$$F_R = \mathbb{E}\{F_R\} = \frac{1}{P_{\text{sys}}} \left(\sum_{g=1}^{G} \frac{\mathcal{P}_g K_g F_g P_g^{\max}}{R_g} \right), \tag{35}$$

$$R_R = \frac{1}{P_{\text{sys}}} \left(\underbrace{\sum_{g=1}^{G} \frac{\mathcal{P}_g K_g P_g^{\max}}{R_g}}_{\mathbb{E}\{R_R\}} + \frac{P^{\text{FESS}}}{R^{\text{FESS}}} \right), \tag{36}$$

$$\xi = 0.5 \times \frac{2H + T_R(D+F_R)}{\sqrt{2HT_R(D+R_R)}}, \tag{37}$$

$$\omega_n = \sqrt{\frac{D+R_R}{2HT_R}}, \quad \phi = \tan^{-1}\left(\frac{\xi}{\sqrt{1-\xi^2}} \right). \tag{38}$$

The maximum frequency deviation (nadir) can be estimated at the first zero crossing of the RoCoF. Hence, taking the derivative of (34) and equating it to zero, the frequency nadir can then be expressed as:

$$\Delta f^{\max} = \frac{\Delta P_L}{D+R_R} \left(1 + e^{-\xi\omega_n t^{\max}} \sqrt{\frac{T_R(R_R - F_R)}{2H}} \right) \tag{39}$$

where

$$t^{\max} = \frac{1}{\omega_n\sqrt{1-\xi^2}} tan^{-1} \left(\frac{\omega_n\sqrt{1-\xi^2}}{\xi\omega_n - 1/T_R} \right) \tag{40}$$

It can be noticed that the frequency nadir depends heavily on the frequency security parameters (D, H, F_R, R_R), all of which depend on the generator synchronization probabilities as seen in equations (32), (33), (35), (36), respectively. Next, the maximum RoCoF is obtained by derivating (34) and setting $t = 0^+$. It is expressed as follows:

$$\left| \frac{df}{dt} \right|^{\max} = \left| \frac{df}{dt} \right| (0^+) = \frac{\Delta P_L}{2H\sqrt{1-\xi^2}} \cos(\phi) = \frac{\Delta P_L}{2H} \tag{41}$$

Similarly, the quasi-steady-state (QSS) frequency deviation can be estimated using the final value theorem as follows:

$$\Delta f^{\text{ss}} = \lim_{s\to 0} s\Delta f(s) = \frac{\Delta P_L}{D+R_R} \tag{42}$$

It can be observed that the maximum RoCoF and QSS frequency are also dependent on the generator synchronization probabilities via the frequency security parameters H, D, F_R, and R_R. The resulting expressions in equations (39), (41), and (42) are used as constraints for the optimization problem that determines the FESS sizes.

B. Inertia and Droop Constants for the FESS

The control strategy for ESS is formulated in this section. The IR support is achieved using a derivative control on the frequency signal, whereas PFR support is provided using droop control on the frequency deviation, as seen in Fig. 3.

Reformulating the swing equation, the inertia constant for the FESS can be expressed as follows [20]:

$$H^{\text{FESS}} = \max(p^{\text{IR}}(t)) \frac{f_R}{2} \left(\left| \frac{df}{dt} \right|^{\max} \right)^{-1} \tag{43}$$

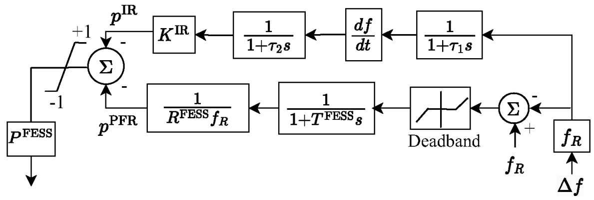


Fig. 3: FESS participating in IR and PFR.

where f_R is the rated frequency and $\left|\frac{df}{dt}\right|^{\max}$ is the maximum RoCoF without FESS support. From Fig. 3, the reference signal for IR is obtained as follows:

$$p^{\mathrm{IR}}(t) = K^{\mathrm{IR}} \frac{1}{(1+s\tau_1)(1+s\tau_2)} \frac{df}{dt}(t) \tag{44}$$

However, when the value of $p^{\mathrm{IR}} \geq 1$, which can occur due to high RoCoF or a high K^{IR} setting, the power capacity constraint limits p^{IR} to 1. In this case, the expression for inertia constant from FESS is expressed as follows:

$$H^{\mathrm{FESS}} = \frac{f_R}{2} \left(\left| \frac{df}{dt} \right|^{\max} \right)^{-1} \tag{45}$$

Hence, ignoring the filtering time constants, the inertia constant for FESS can be approximated as follows:

$$H^{\mathrm{FESS}} = \begin{cases} K^{\mathrm{IR}} \frac{f_R}{2} & \text{if } |p^{\mathrm{IR}}| \leq 1 \\ \frac{f_R}{2} \left(\left| \frac{df}{dt} \right|^{\max} \right)^{-1} & \text{if } |p^{\mathrm{IR}}| > 1 \end{cases} \tag{46}$$

The PFR support from the FESS is achieved using the PFR control loop illustrated in Fig. 3. The value of R^{FESS} is determined using the desired QSS frequency and the dead band setting. After the dead band, a constant droop value provides PFR until $p^{\mathrm{PFR}} \leq 1$. The effective droop then assumes a linear form for larger frequency deviations since p^{PFR} saturates to 1 p.u. The ESS droop constant can be expressed as follows:

$$R^{\mathrm{FESS}} = \begin{cases} \frac{1}{f_R} \left(\Delta f^{\mathrm{ss,des}} - \Delta f^{\mathrm{db}} \right) & \text{if } \Delta f^{\mathrm{ss}} \leq \Delta f^{\mathrm{ss,des}} \\ \Delta f^{\mathrm{ss}} - \frac{1}{f_R} \Delta f^{\mathrm{db}} & \text{if } \Delta f^{\mathrm{ss}} > \Delta f^{\mathrm{ss,des}} \end{cases} \tag{47}$$

where $\Delta f^{\mathrm{ss,des}}$ is the desired QSS frequency setting, Δf^{db} is the deadband setting, and Δf^{ss} is the actual QSS frequency.

C. Optimization Framework for Sizing of ESSs

The optimal FESS sizes for IR and PFR are obtained with a non-linear optimization problem. The objective function of the formulation is to minimize the total energy rating of the FESS. The formulation is solved considering the frequency security and SoC constraints. The problem is structured as follows:

$$\min_{P^{\mathrm{FESS}}} \quad S^{\mathrm{FESS}} \tag{48a}$$

subject to

$$\frac{\Delta P_L}{2H} \leq \left| \frac{df}{dt} \right|^{\mathrm{lim}} \tag{48b}$$

$$\frac{\Delta P_L}{R_R + D} \leq \Delta f_{\mathrm{ss}}^{\mathrm{lim}} \tag{48c}$$

$$\frac{\Delta P_L}{R_R + D} e^{-\xi \omega_n t^{\max}} \sqrt{\frac{T_R(R_R - F_R)}{2H}} \leq \Delta f^{\mathrm{lim}} \tag{48d}$$

$$0.5 S^{\mathrm{FESS}} - \frac{P^{\mathrm{FESS}} t^{\mathrm{req}}}{\eta^-} \geq 0.1 S^{\mathrm{FESS}} \tag{48e}$$

$$0.5 S^{\mathrm{FESS}} + P^{\mathrm{FESS}} t^{\mathrm{req}} \eta^+ \leq 0.9 S^{\mathrm{FESS}} \tag{48f}$$

where ΔP is the disturbance size, $\left|\frac{df}{dt}\right|^{\mathrm{lim}}$, $\Delta f_{\mathrm{lim}}^{\mathrm{ss}}$, Δf^{lim} are the maximum RoCoF, QSS frequency, and frequency nadir safe limits respectively, t^{req} is the required time for frequency support. Constraints (48b)-(48d) represent the frequency security constraints while (48e) and (48f) are the FESS SoC constraints for under and over-frequency events respectively.

VI. Simulation and Results

Simulations are performed on the IEEE RTS-GMLC test system to demonstrate the proposed approach. Figure 4 illustrates the single line diagram of the IEEE RTS-GMLC system. The GMLC system is a modification of the original IEEE-RTS system, and its details are present in [32]. The RTS-GMLC introduces RERs consisting of PV, concentrated solar power (CSP), hydro, wind, and storage units. The RERs resource data database consists of day-ahead forecasts recorded at hourly intervals for a year, which is used for all PV farms in this work. The wind farm outputs in the system are remodeled using the procedure in section II to account for the effects of stochasticity and geographical dispersion on ESS sizes. The RTS-GLMC has 73 nodes and 120 transmission lines. The generation mix includes 74 conventional generation units, 20 hydro units, 25 PV farms, 31 roof-top PV (RTPV), 1 CSP, 4 winds farms, and 1 ESS. The ESS provided by the GMLC is not considered in this work since we are concerned with sizing the ESS for this system. Each region consists of distributed loads with a peak load of 2850 MW per region. Since the combined generation of the system is significantly larger than the system peak load, the system is overbuilt and very reliable [32]. Hence, in this work, the demand is scaled by 1.2 to increase the load curtailment rate and loading of transmission lines. In addition, to improve the computational efficiency of the proposed approach, a multi-region transportation model of the RTS-GMLC system is used for composite reliability analysis and ESS sizing. Readers are referred to [28] for the details about constructing the transportation models for reliability analysis. The MCS framework proposed in this work is simulated in Python with the DC-OPF solved using the *glpk* solver in *pyomo* [33]. The non-linear optimization framework in section V-C is solved in MATLAB using the *fmincon* solver.

Wind speeds are extracted from the National Renewable Energy Laboratory (NREL) resource extraction tool [34] for Fort Collins, Colorado (CO), Colorado Springs, CO, and Boise, Idaho from 2007-2014. The input wind speeds are fuzzy C-means clustered to obtain 12 discrete wind speed states per location, which are then used to build the state transition matrices using the procedure in section II. The statistical details for the three locations are present in Table I. The values in the table show that the wind power between Fort Collins and Colorado Springs has a larger correlation due to the smaller distance. They also have higher average wind speed than Boise, Idaho's wind profile. The varying wind speeds and correlation levels will help to illustrate the impacts of siting wind farms in these locations on the sizing of RESS and FESS.

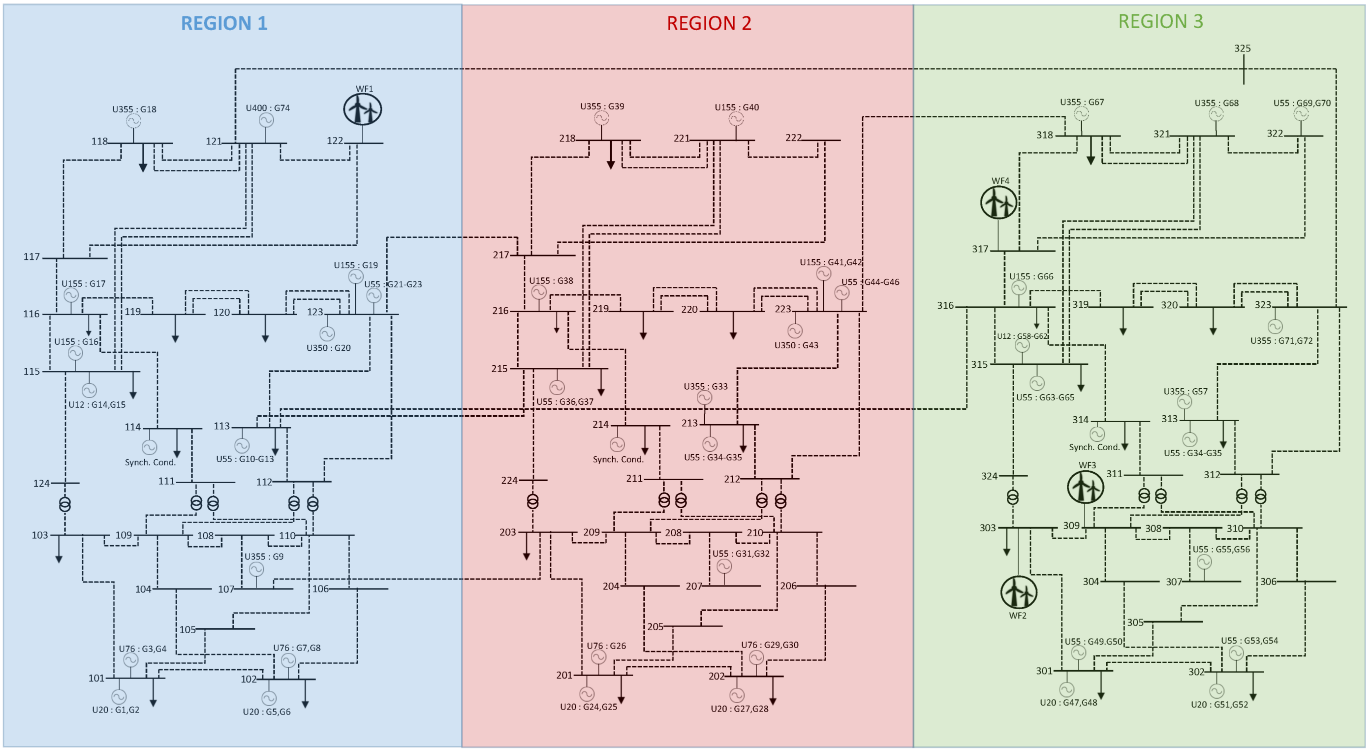


Fig. 4: Single line diagram of the IEEE RTS-GMLC test system with conventional generators and wind farm positions [31].

TABLE I: Statistical parameters for wind speeds.

Location	Mean Speed (m/s)	Variance (m/s^2)	Correlation	Distance (miles)
Fort Collins	4.4994	13.537	$--$	$--$
CO Springs	5.0300	14.711	0.4463	133.2
Boise	4.3476	11.420	0.0338	749.6

Since the proposed approach is studied for systems with high wind penetration, the simulation scenarios are formulated considering uncertainty in wind speed based on the location of wind farms. The simulation scenarios are as follows:

- Case I, Case III, and Case III: These cases consider wind speeds of a single location for all wind farms in the test system. For case I, Fort Collins is chosen as the location for all the wind farms. For cases II and III, CO Springs and Boise are the locations for all wind farms respectively.
- Case IV: In this case, the wind farm in region 1 (WF1 in Fig. 4) considers the wind speeds of CO Springs, while the wind farms in region 3 (WF2, WF3, and WF4) consider wind speeds of Fort Collins. This case illustrates the effects of the geographical dispersion of wind farms on the RESS and FESS sizing.
- Case V: For case V, the WF1 is subject to wind speeds of Boise, while the WF2, WF3, and WF4 are subject to wind speeds of Fort Collins. This case considers the effect of wind speed correlation on the sizing of RESS and FESS.

For each simulation scenario, the sequential MCS in section III is performed and the reliability indices and generator synchronization probabilities are obtained. The resulting values from the MCS are then used to determine ESS sizes. The results of the proposed approach are highlighted as follows:

A. *Composite Reliability Evaluation and Sizing of RESS*

The system composite reliability assessment results are summarized in Table II. From the table, the following can be observed:

- Initially, cases I, II, and III show that the location mean wind speeds significantly affect system reliability for non-distributed wind farms. Case II has the smallest value for reliability indices because CO Springs has a higher mean wind speed than other locations. It can also be noticed that case III has slightly smaller reliability indices than case I despite Boise having a smaller mean wind speed. This can be attributed to a smaller variance in Boise wind compared to Fort Collins.
- Next, the reliability indices for cases IV and V illustrate the benefit of the geographical dispersion of wind power. It can be observed that both these cases with wind power dispersion have smaller reliability indices than all previous cases. A major advantage of wind power dispersion can be observed by comparing cases II and V, where despite dispersing the wind power between lower speed sites (Boise and Fort Collins) in case V, the reliability indices are smaller than in case II. Between cases IV and V, case IV yields a smaller value for indices due to the large difference in mean wind speed between Boise and CO Springs.

TABLE II: Composite reliability analysis of the RTS GMLC for the proposed cases.

Cases	LOLP	LOLE (h/yr)	LOLF (f/yr)	EDNS (MW/yr)	EUE (MWh/yr)
Case I	0.001887	16.5283	6.4204	0.4396	3850.6
Case II	0.001771	15.5177	6.1137	0.4137	3624.1
Case III	0.001885	16.5149	6.2612	0.4379	3836.1
Case IV	0.001646	14.4147	5.7337	0.3784	3315.1
Case V	0.001730	15.1525	5.9191	0.3964	3472.7

In this paper, the target loss of load probability (LOLP) for RESS sizing is set as 0.000274, which is derived from the "one day in 10 years" criterion in section A.3 of the NERC standard BAL-502-RF-03 [35]. The composite reliability analysis in Table II clearly shows that all the cases have LOLP significantly greater than the NERC limit. Hence, the RESS are sized for all the cases, and their ratings are summarized in Table III. The unavailability ratio α is initially calculated using the ratio of NERC unavailability limit and the LOLP values in Table II. Then, using equation (25), the time rating of the RESS t^{RESS} is determined. The power ratings for the RESS are obtained using equation (19). The following can be observed from Table III:

- The power ratings for the RESS in cases I and II are proportional to the reliability indices (case II has a smaller ESS rating since it has a lower value for reliability indices). However, in case III, the largest values for the RESS ratings are observed despite case III having smaller values for reliability indices than case I. It should be noted that the power rating of the RESS is dependent on the mean value of load curtailment, which can have a high value despite low values for LOLP, LOLF, and EDNS. It can also be observed that case III has the largest energy rating due to a considerably high MDT.
- The benefits of wind power dispersion are evident for the RESS ratings as cases IV and V have smaller RESS sizes than cases I, II, and III. The smallest RESS ratings are observed for case IV since it has the lowest load curtailment rate and MDT and is closely followed by case V.

TABLE III: RESS sizes for the proposed cases.

Cases	MDT (h)	α	t^{RESS} (h)	P^{RESS} (MW)	S^{RESS} (MWh)
Case I	2.5744	0.1452	5.2290	551.65	2884.5
Case II	2.5382	0.1547	4.9870	543.85	2712.1
Case III	2.6377	0.1453	5.3552	562.47	3012.7
Case IV	2.5140	0.1665	4.7444	527.87	2504.4
Case V	2.5599	0.1584	4.9653	535.05	2656.7

From the results, it is evident that the wind farm locations directly impact system reliability and RESS sizes since the locations influence the capacity value of wind farms. The proposed approach for wind stochasticity modeling using transition rate matrices within the MCS captures the aforementioned characteristic. The results also show that the dispersion of wind power significantly benefits system reliability and lowers the RESS size requirements. Despite dispersing the wind power between locations with lower mean wind speeds, considerable improvement in system reliability and reduction in RESS sizes are observed.

B. Sizing of the Frequency Security ESS

The FESS are sized based on the system frequency security requirements. In this work, we assume the frequency support from ESS is provided using the standard battery-inverter-grid topology. More details on the construction of ESS providing frequency response can be found in [36]. Initially, we present a simulation-based procedure for selecting the virtual inertia gain and droop constant. Then, the FESS power and energy ratings are estimated using these values and the probability of generator synchronization obtained from the MCS.

1) Selection of IR gain and droop constant for the frequency security ESS: Along with the FESS power and energy rating, it is important to select appropriate values for the inertia gain factor K^{IR} and droop constant R^{FESS}. Including these parameters in the sizing algorithm in equation (48) is infeasible as these parameters introduce constraints that can only be enforced with dynamic simulations. Hence, in this paper, a simulation-based approach is used to determine the values of K^{IR} and R^{FESS}. The circuit in Fig. 5 is simulated in MATLAB Simulink to obtain the system frequency response with the frequency support of a FESS. The simulation circuit is constructed by combining the IR and PFR control loops of the FESS in Fig. 3 with the aggregated system LFC model. The virtual inertia gain and droop constant are determined using the frequency response of Fig. 5 under a step disturbance.

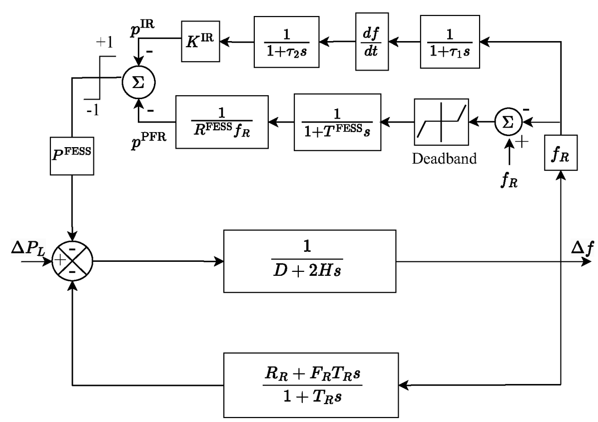
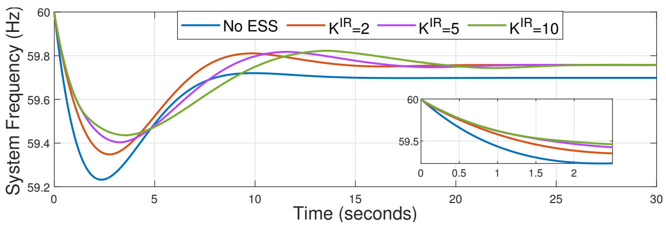
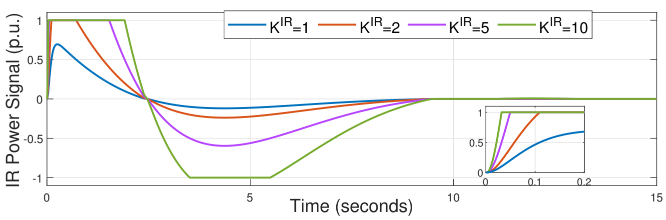


Fig. 5: Aggregated LFC model with FESS integration.

The frequency response obtained by simulating Fig. 5 is displayed in Fig. 6. By selecting a larger IR gain parameter K^{IR}, FESS can provide a fast response to the change in system RoCoF. Consequently, a larger FESS inertia constant can be observed with a larger gain. This effect is observed in Fig. 6a where the frequency nadir reduces as IR gain increases. However, it is important to consider the effect of a larger gain after the first zero crossing of the RoCoF. After the zero crossing, the FESS switch to charging mode (negative IR power signal) due to RoCoF reversal as seen in Fig. 6b. A high K^{IR} during this stage causes the IR signal to interfere with the PFR signal, which increases the settling time for frequency deviation as seen in Fig. 6a. In addition, the improvement in system RoCoF saturates for larger IR gain values, as seen in the zoomed-in plot in Fig. 6a. On the contrary, selecting a small value of IR gain can lead to inadequate IR response as seen in Fig. 6b for $K^{\mathrm{IR}} = 1$. Hence, in this work, $K^{\mathrm{IR}} = 5$ is chosen as it can provide a reasonable response without having a high charging characteristic during frequency recovery.



(a)



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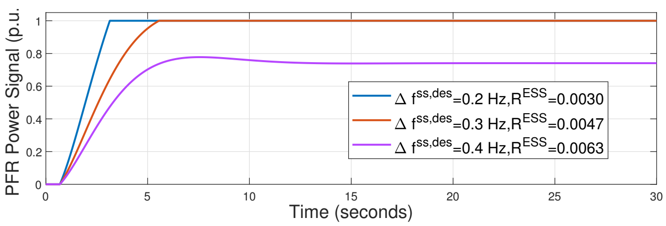

(c)

Fig. 6: a) System frequency deviation b) FESS IR loop response and, c) PFR during a disturbance.

Similarly, for the selection of the droop constant, it can be seen in Fig. 6c that a smaller droop can provide a better response. The droop coefficient of 0.003 renders the FESS to generate 1 p.u. the fastest during the QSS frequency, whereas for droop of 0.0063, FESS generates around 0.75 p.u. However, selecting a small droop value can lead to overestimating the contribution of FESS to R_R in the LFC model. Hence, for better generalization, the droop constant is calculated by setting the desired QSS frequency $\Delta f^{\mathrm{ss,des}}$ equal to the QSS frequency safe limit $\Delta f_{\mathrm{ss}}^{\mathrm{lim}}$ in equation (47). This ensures that the FESS generates 1 p.u. when the QSS frequency is greater than or equal to the safe limit.

2) Sizing results for the frequency security ESS: The probability of synchronization obtained using the MCS and the methods in section V are deployed to obtain the sizes for the FESS. The safety limits and disturbance sizes to size the FESS are set according to the prevailing limits in the Eastern Interconnection (EI), Western Interconnection (WI), and ERCOT in the United States, found in Table 1 of the NERC standard BAL-003-1 [37]. However, since the EI and WI have the same under-frequency load-shedding (UFLS) setting, only the frequency security limits of EI are considered among these two interconnections. The values of frequency security limits in Table IV and disturbance size considered in this work are obtained as follows:

- Disturbance size (ΔP_L): The resource contingency criterion (RCC) set forth by the NERC in standard BAL-003-1 specifies an N-2 criterion for the disturbance. Hence, in this paper, the disturbance size is taken as the failure of two largest generators G74 (400 MW) and G33 (355 MW).
- Frequency nadir limits (Δf^{lim}): They are set according to the prevailing UFLS limits in NERC standard BAL-003-1.
- RoCof limits ($\left|\frac{df}{dt}\right|^{\mathrm{lim}}$): They are calculated using the lowest system inertia experienced by the interconnections in 2017 [38] and 10 % of the system demand during that time [39].
- QSS frequency limits ($\Delta f_{\mathrm{ss}}^{\mathrm{lim}}$): They are set as 50% of the frequency nadir limits.

TABLE IV: Frequency security limits.

Interconnection	Nadir limit (Hz)	RoCoF limit (Hz/s)	QSS frequency limit (Hz)
Eastern	0.5	0.5124	0.25
ERCOT	0.7	0.6608	0.35

Table V illustrates the FESS sizing results for system frequency security based on the limits described earlier. The following can be observed from Table V:

- Initially, comparing cases I, II, and III, it can be observed that the expected values of system inertia are inversely proportional to the mean locational wind speeds and variance. For example, case II has the lowest values for inertia since CO Springs has the highest mean wind speed. This is due to the displacement of conventional generators by wind farms. This can also be observed in Fig. 7, where the conventional generators have a lower probability of synchronization in case II compared to cases I and III. As a result, case II requires the largest FESS sizes, while case III has the lowest FESS requirement for both EI and ERCOT limits.
- The reliability simulations in the last section showed that cases IV and V have higher capacity values of wind farms. Despite the increased capacity values in these cases, the FESS sizes do not exceed those in case II. It is because along with the reduction of global wind variance, the dispersion of wind farms ensures that the extreme locational wind speeds are flattened. For example, in case II, if there is high wind speed in CO Springs, all wind farms experience this speed. This causes maximum displacement of generators by wind farms during dispatch. However, high wind speed at CO Springs in case IV means only WF1 experiences this speed. This causes the generators to maintain a higher synchronization probability, as seen in Fig. 7 for cases IV and V. Consequently, the FESS size requirements reduce with the dispersion of wind farms.

C. Deployment of RESS and FESS in the RTS-GMLC System

Initially, the RESS sizes obtained for case IV in Table III are deployed in the RTS-GMLC system using the RESS deployment strategy in section IV-B. Since case IV has the lowest RESS size requirement, this case is chosen for RESS

TABLE V: Expected system inertia and FESS sizing results. (inertia is normalized to a system base of 100 MVA).

Cases	$\mathbb{E}\{H\}$ s	EI limits		ERCOT limits	
		P^{FESS} (MW)	S^{FESS} (MWh)	P^{FESS} (MW)	S^{FESS} (MWh)
Case I	175.448	1345.5	989.31	819.20	602.35
Case II	173.618	1361.7	1001.2	835.07	614.02
Case III	176.650	1334.6	981.32	808.45	594.45
Case IV	174.713	1351.4	993.59	824.68	606.38
Case V	175.643	1343.1	987.59	816.72	600.53

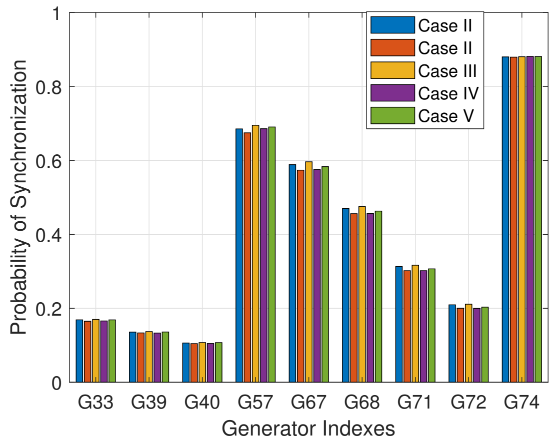

Fig. 7: Synchronization probability of RTS-GMLC generators with the highest inertia.

deployment. The improvement in the system reliability with the proposed 527.87 MW, 2504.4 MWh RESS is illustrated with the probability distribution of loss of load hours in Fig. 8. As expected, when the proposed RESS is deployed, the system loss of load hours are lower than the NERC reliability limit in the majority of Monte Carlo samples.

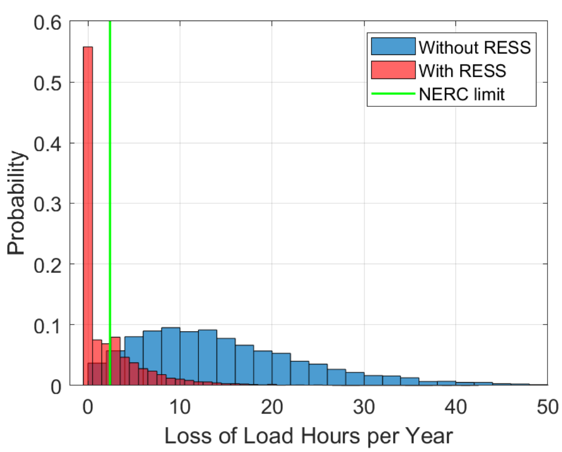

Fig. 8: Probability distribution of system loss of load hours.

To illustrate the versatility of the proposed RESS sizing method, the RESS are sized and deployed for varying levels of system demand scaling as illustrated in Table VI. From the table, it is clear that as demand scaling increases, the reliability indices worsen significantly. As a result, larger RESS sizes are required to ensure system reliability. However, when the suggested RESS are deployed in the system, the reliability indices for all three demand-scaled cases improve and are within the NERC unavailability limit (LOLP) of 0.000274. In addition, it can be noticed that the proposed RESS maintain the system LOLP very close to the limit for all three cases. This shows that the proposed approach is economically feasible as it does not significantly overestimate the RESS sizes.

Next, the sized FESS in case IV is deployed into the RTS-GMLC system using the values of inertia gain and droop constant obtained from section VI-B1 and the power and energy rating obtained from Table V. The results of the Monte Carlo simulation with the deployment of case IV FESS are illustrated with the cumulative distributions of frequency nadir in Fig. 9. The figure shows that without FESSs, the system frequency nadir has very high values. This is due to the N-2 criterion system disturbance and lack of system inertia and regulation with high wind penetration. The expected value of the frequency nadir for the case without FESS is observed to be 1.2644 Hz, which is considerably larger than the EI and ERCOT nadir limits. For the cases where EI and ERCOT FESS are deployed (red and yellow curves), the expected values of frequency nadir are 0.5269 and 0.7442, respectively. These values are close to the EI and ERCOT nadir limits of 0.5 Hz and 0.7 Hz, respectively. If a higher level of frequency security is desired, the planners can use smaller values of inertia (H) and other parameters (D, F_R, R_R) from their probability distributions instead of the expected value (example: first quartile values). Similar to the RESS sizes, the suggested FESS sizes successfully keep the system frequency nadir close to the limits for which the FESS are sized, which displays the economical feasibility of the proposed approach.

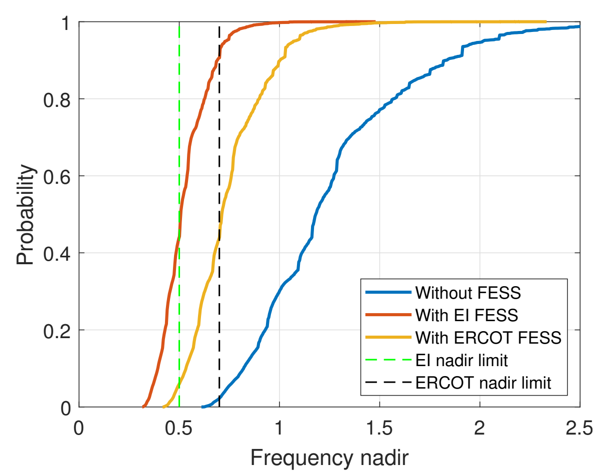

Fig. 9: Cumulative distribution of system frequency nadir.

D. Discussion

Simulation results show that the proposed approach can simultaneously size the ESS for reliability and frequency security while incorporating wind stochasticity, geographical characteristics of wind farms, forced outages of components, and frequency support from ESSs. For each simulation scenario, it can be seen that a single MCS is adequate for generating the composite reliability indices for sizing the RESS and obtaining

TABLE VI: IEEE RTS-GMLC reliability indices with and without RESS deployment.

Demand scaling factor	Reliability indices without RESS					RESS Size		Reliability indices with RESS				
	LOLP	LOLE (h/yr)	LOLF (f/yr)	EDNS (MW/yr)	EUE (MWh/yr)	P^{RESS} (MW)	S^{RESS} (MWh)	LOLP	LOLE (h/yr)	LOLF (f/yr)	EDNS (MW/yr)	EUE (MWh/yr)
1.15	0.00038	3.3346	1.4904	0.0073	641.91	377.17	292.16	0.00025	2.2082	0.9123	0.0053	464.92
1.20	0.00164	14.415	5.7337	0.3784	3315.1	532.26	2547.3	0.00025	2.1622	0.8021	0.0064	561.13
1.25	0.00570	49.919	16.128	1.6614	14554	877.43	8676.1	0.00023	2.0440	0.8243	0.0045	391.39

the synchronization probability of conventional generators for FESS sizing. The method is versatile as it is easily generalizable for varying reliability levels and frequency security limits, which is illustrated with the deployment of ESSs. The results also illustrate the benefits of the geographical dispersion of wind farms on ESS sizing as follows:

- Improvement in system reliability indices and reduction of RESS sizes by reducing the global wind variance and increasing wind power capacity value.
- Reduction in FESS sizes by improving the synchronization probability of conventional generators. This benefit is achieved by alleviating the conventional generator displacement due to extreme wind speeds in a single location.

VII. Conclusion

This paper proposes a probabilistic approach for sizing ESS to enhance system reliability and frequency security of wind-rich grids. A sequential Monte Carlo simulation framework is proposed that generates reliability indices for the sizing of reliability ESS and generator synchronization probabilities for frequency support ESS. Case studies on the IEEE RTS-GMLC show that the proposed approach generates optimal ESS sizes while incorporating uncertainties in wind generation, availability of system components, and operation constraints. The results also illustrate the benefits of wind farm dispersion in improving the system reliability and reducing the sizing requirements for the ESS. The proposed approach offers flexibility for system planners as the sizing algorithms can be incorporated with the existing composite reliability assessment frameworks. In addition, the ESS sizes can easily be determined for varying levels of desired reliability and frequency security. Future work involves improving the ESS sizing algorithms to incorporate complex control architectures and different storage alternatives.

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References

[1] D. Pandit, A. Bera, R. K. Saket, J. Mitra, and N. Nguyen, "Energy storage and geographical distribution of wind power to improve frequency stability," in *2023 IEEE IAS Global Conf. Renewable Energy Hydrogen Technol. (GlobConHT)*, pp. 1–6, 2023.

[2] IEA (2022), "Tracking clean energy progress 2023, IEA, Paris." https://www.iea.org/reports/tracking-clean-energy-progress-2023, License: CC BY 4.0v. Online; Accessed 20 Jun. 2023.

[3] D. Pandit, "Reliability and stability assessment of renewable energy and energy storage integrated systems," Master's thesis, University of Wyoming, 2022.

[4] Y.-K. Wu, S.-M. Chang, and P. Mandal, "Grid-connected wind power plants: A survey on the integration requirements in modern grid codes," *IEEE Trans. Ind. Appl.*, vol. 55, no. 6, pp. 5584–5593, 2019.

[5] N. Nguyen, A. Bera, and J. Mitra, "Energy storage to improve reliability of wind integrated systems under frequency security constraint," *IEEE Trans. Ind. Appl.*, vol. 54, no. 5, pp. 4039–4047, 2018.

[6] K. Ullah, Z. Ullah, S. Aslam, M. S. Salam, M. A. Salahuddin, M. F. Umer, M. Humayon, and H. Shaheer, "Wind farms and flexible loads contribution in automatic generation control: An extensive review and simulation," *Energies*, vol. 16, no. 14, p. 5498, 2023.

[7] M. Gholami, S. Mousavi, and S. Muyeen, "Enhanced microgrid reliability through optimal battery energy storage system type and sizing," *IEEE Access*, 2023.

[8] P. P. Gupta, P. Jain, S. Sharma, and R. Bhakar, "Reliability-security constrained unit commitment based on benders decomposition and mixed integer non-linear programming," in *2017 Int. Conf. on Computer, Commun. and Electron. (Comptelix)*, pp. 328–333, IEEE, 2017.

[9] H. Xie, X. Teng, Y. Xu, and Y. Wang, "Optimal energy storage sizing for networked microgrids considering reliability and resilience," *IEEE Access*, vol. 7, pp. 86336–86348, 2019.

[10] S. Bahramirad, W. Reder, and A. Khodaei, "Reliability-constrained optimal sizing of energy storage system in a microgrid," *IEEE Trans. Smart Grid*, vol. 3, no. 4, pp. 2056–2062, 2012.

[11] K. Lai and L. Zhang, "Sizing and siting of energy storage systems in a military-based vehicle-to-grid microgrid," *IEEE Trans. Ind. Appl.*, vol. 57, no. 3, pp. 1909–1919, 2021.

[12] X. Zhang and A. J. Conejo, "Coordinated investment in transmission and storage systems representing long- and short-term uncertainty," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 7143–7151, 2018.

[13] Y. Li, B. Feng, B. Wang, and S. Sun, "Joint planning of distributed generations and energy storage in active distribution networks: A bi-level programming approach," *Energy*, vol. 245, p. 123226, 2022.

[14] S. Sulaeman, Y. Tian, M. Benidris, and J. Mitra, "Quantification of storage necessary to firm up wind generation," *IEEE Trans. Ind. Appl.*, vol. 53, no. 4, pp. 3228–3236, 2017.

[15] A. Bera, N. Nguyen, S. Alzahrani, K. Sinjari, and J. Mitra, "Variability reduction of wind power using aggregation and energy storage," in *2020 Int. Conf. Probabilistic Methods Appl. to Power Syst. (PMAPS)*, pp. 1–6, 2020.

[16] R. Engleitner, A. Nied, M. S. M. Cavalca, and J. P. da Costa, "Dynamic analysis of small wind turbines frequency support capability in a low-power wind-diesel microgrid," *IEEE Trans. Ind. Appl.*, vol. 54, no. 1, pp. 102–111, 2018.

[17] Z. Zhang, M. Zhou, Z. Wu, S. Liu, Z. Guo, and G. Li, "A frequency security constrained scheduling approach considering wind farm providing frequency support and reserve," *IEEE Trans. Sustain. Energy*, vol. 13, no. 2, pp. 1086–1100, 2022.

[18] A. Bera, N. Nguyen, B. Chalamala, and J. Mitra, "Quantification of storage required for preserving frequency security in wind-integrated systems," *IET Renewable Power Gener.*, vol. 17, no. 9, pp. 2366–2378, 2023.

[19] U. Tamrakar, D. Shrestha, M. Maharjan, B. P. Bhattarai, T. M. Hansen, and R. Tonkoski, "Virtual inertia: Current trends and future directions," *Appl. Sci.*, vol. 7, no. 7, p. 654, 2017.

[20] V. Knap, S. K. Chaudhary, D.-I. Stroe, M. Swierczynski, B.-I. Craciun, and R. Teodorescu, "Sizing of an energy storage system for grid inertial response and primary frequency reserve," *IEEE Trans. Power Syst.*, vol. 31, no. 5, pp. 3447–3456, 2015.

[21] H. Liu, C. Zhang, X. Peng, and S. Zhang, "Configuration of an energy storage system for primary frequency reserve and inertia response of the power grid," *IEEE Access*, vol. 9, pp. 41965–41975, 2021.

[22] A. Bera, B. R. Chalamala, R. H. Byrne, and J. Mitra, "Sizing of energy storage for grid inertial support in presence of renewable energy," *IEEE Trans. Power Syst.*, vol. 37, no. 5, pp. 3769–3778, 2022.

[23] H. Golpîra, A. Atarodi, S. Amini, A. R. Messina, B. Francois, and H. Bevrani, "Optimal energy storage system-based virtual inertia placement: A frequency stability point of view," *IEEE Trans. Power Syst.*, vol. 35, no. 6, pp. 4824–4835, 2020.

[24] K. Brokish and J. Kirtley, "Pitfalls of modeling wind power using markov chains," in *2009 IEEE/PES Power Syst. Conf. and Expo.*, pp. 1–6, IEEE, 2009.

[25] Y. Degeilh and C. Singh, "A quantitative approach to wind farm diversification and reliability," *Int. J. Elect. Power Energy Syst.*, vol. 33, no. 2, pp. 303–314, 2011.

[26] P. Potisomporn and C. R. Vogel, "Spatial and temporal variability characteristics of offshore wind energy in the United Kingdom," *Wind Energy*, vol. 25, no. 3, pp. 537–552, 2022.

[27] M. R. Milligan and T. Factor, "Optimizing the geographic distribution of wind plants in Iowa for maximum economic benefit and reliability," *Wind Engineering*, vol. 24, no. 4, pp. 271–290, 2000.

[28] C. Singh, P. Jirutitijaroen, and J. Mitra, *Electric Power Grid Reliability Evaluation: Models and Methods*. John Wiley & Sons, 2018.

[29] J. Mitra, "Reliability-based sizing of backup storage," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 1198–1199, 2010.

[30] P. M. Anderson and M. Mirheydar, "A low-order system frequency response model," *IEEE Trans. Power Syst.*, vol. 5, no. 3, pp. 720–729, 1990.

[31] A. Mate and E. Cotilla-Sanchez, "Rapid method for generation prioritization during system restoration with renewable resources," in *2019 IEEE/IAS 55th Ind. Commercial Power Syst. Tech. Conf. (I&CPS)*, pp. 1–8, 2019.

[32] C. Barrows, A. Bloom, A. Ehlen, J. Ikäheimo, J. Jorgenson, D. Krishnamurthy, J. Lau, B. McBennett, M. O'Connell, E. Preston, A. Staid, G. Stephen, and J.-P. Watson, "The ieee reliability test system: A proposed 2019 update," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 119–127, 2020.

[33] M. L. Bynum, G. A. Hackebeil, W. E. Hart, C. D. Laird, B. L. Nicholson, J. D. Siirola, J.-P. Watson, and D. L. Woodruff, *Pyomo–optimization modeling in python*, vol. 67. Springer Science & Business Media, third ed., 2021.

[34] M. Rossol and G. Buster, "NREL/rex: TemporalStats move." https://doi.org/10.5281/zenodo.4499033, Feb. 2021.

[35] North American Electric Reliability Corporation, "NERC Reliability Standard Standard BAL-502-RF-03: Planning Resource Adequacy Analysis, Assessment and Documentation." Atlanta, GA.

[36] D. Greenwood, K. Y. Lim, C. Patsios, P. Lyons, Y. S. Lim, and P. Taylor, "Frequency response services designed for energy storage," *Appl. Energy*, vol. 203, pp. 115–127, 2017.

[37] North American Electric Reliability Corporation, "NERC Reliability Standard BAL-003-1: Frequency Response and Frequency Bias Setting." Atlanta, GA.

[38] North American Electric Reliability Corporation, "Forward Looking Frequency Trends Technical Brief," March 2018. [Online]. Available: https://www.nerc.com/comm/Other/essntlrlbltysrvcstskfrcDL/ERS_Forward_Measures_124_Tech_Brief_03292018_Final.pdf.

[39] H. Ahmadi and H. Ghasemi, "Security-constrained unit commitment with linearized system frequency limit constraints," *IEEE Trans. Power Syst.*, vol. 29, no. 4, pp. 1536–1545, 2014.

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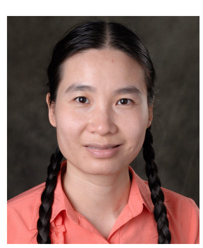

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